Liquidity Monitoring and Measurement

Stress Testing

Liquidity stress testing is performed for each of Citi's major entities, operating subsidiaries and/or countries. Stress testing and scenario analyses are intended to quantify the potential impact of a liquidity event on the balance sheet and liquidity position, and to identify viable funding alternatives that can be utilized. These scenarios include assumptions about significant changes in key funding sources, market triggers (such as credit ratings), potential uses of funding and political and economic conditions in certain countries. These conditions include expected and stressed market conditions as well as Company-specific events.

Liquidity stress tests are conducted to ascertain potential mismatches between liquidity sources and uses over a variety of time horizons (overnight, one week, two weeks, one month, three months, one year) and over a variety of stressed conditions. Liquidity limits are set accordingly. To monitor the liquidity of an entity, these stress tests and potential mismatches are calculated with varying frequencies, with several tests performed daily.

Given the range of potential stresses, Citi maintains a series of contingency funding plans on a consolidated basis and for individual entities. These plans specify a wide range of readily available actions for a variety of adverse market conditions or idiosyncratic stresses.

Short-Term Liquidity Measurement: Liquidity Coverage Ratio (LCR)

In addition to internal measures that Citi has developed for a 30-day stress scenario, Citi also monitors its liquidity by reference to the LCR, as calculated pursuant to the U.S. LCR rules.

Generally, the LCR is designed to ensure that banks maintain an adequate level of HQLA to meet liquidity needs under an acute 30-day stress scenario. The LCR is calculated by dividing HQLA by estimated net outflows over a stressed 30-day period, with the net outflows determined by applying prescribed outflow factors to various categories of liabilities, such as deposits, unsecured and secured wholesale borrowings, unused lending commitments and derivatives-related exposures, partially offset by inflows from assets maturing within 30 days. Banks are required to calculate an add-on to address potential maturity mismatches between contractual cash outflows and inflows within the 30-day period in determining the total amount of net outflows. The minimum LCR requirement is 100%, effective January 2017.

In December 2016, the Federal Reserve Board adopted final rules which require additional disclosures relating to the LCR of large financial institutions, including Citi. Among other things, the final rules require Citi to disclose components of its average HQLA, LCR and inflows and outflows each quarter. In addition, the final rules require disclosure of Citi's calculation of the maturity mismatch add-on as well as other qualitative disclosures. The effective date for these disclosures is April 1, 2017.

The table below sets forth the components of Citi's LCR calculation and HQLA in excess of net outflows for the periods indicated:

In billions of dollars	**Dec. 31, 2016**	Sept. 30, 2016	Dec. 31, 2015
HQLA	**$403.7**	$ 403.8	$ 389.2
Net outflows	**332.5**	335.3	344.4
LCR	**121%**	120%	113%
HQLA in excess of net outflows	**$71.3**	$ 68.5	$ 44.8

Note: Amounts set forth in the table above are presented on an average basis.

As set forth in the table above, Citi's LCR increased both year-over-year and sequentially. The increase year-over-year was driven by both an increase in HQLA and a reduction in net outflows. Sequentially, the increase was driven by a slight reduction in net outflows, as HQLA remained largely unchanged.

Long-Term Liquidity Measurement: Net Stable Funding Ratio (NSFR)

In the second quarter of 2016, the Federal Reserve Board, the FDIC and the OCC issued a proposed rule to implement the Basel III NSFR requirement.

The U.S.-proposed NSFR is largely consistent with the Basel Committee's final NSFR rules. In general, the NSFR assesses the availability of a bank's stable funding against a required level. A bank's available stable funding would include portions of equity, deposits and long-term debt, while its required stable funding would be based on the liquidity characteristics of its assets, derivatives and commitments. Standardized weightings would be required to be applied to the various asset and liabilities classes. The ratio of available stable funding to required stable funding would be required to be greater than 100%. While Citi believes that it is compliant with the proposed U.S. NSFR rules as of December 31, 2016, it will need to evaluate any final version of the rules, which are expected to be released during 2017. The proposed rules would require full implementation of the U.S. NSFR beginning January 1, 2018.